Lecture 17

Ray theory

From last time…

Plane wave ansatz:

$$u_i(\mathbf{x}, t) = f(t - p_j x_j) a_i.$$

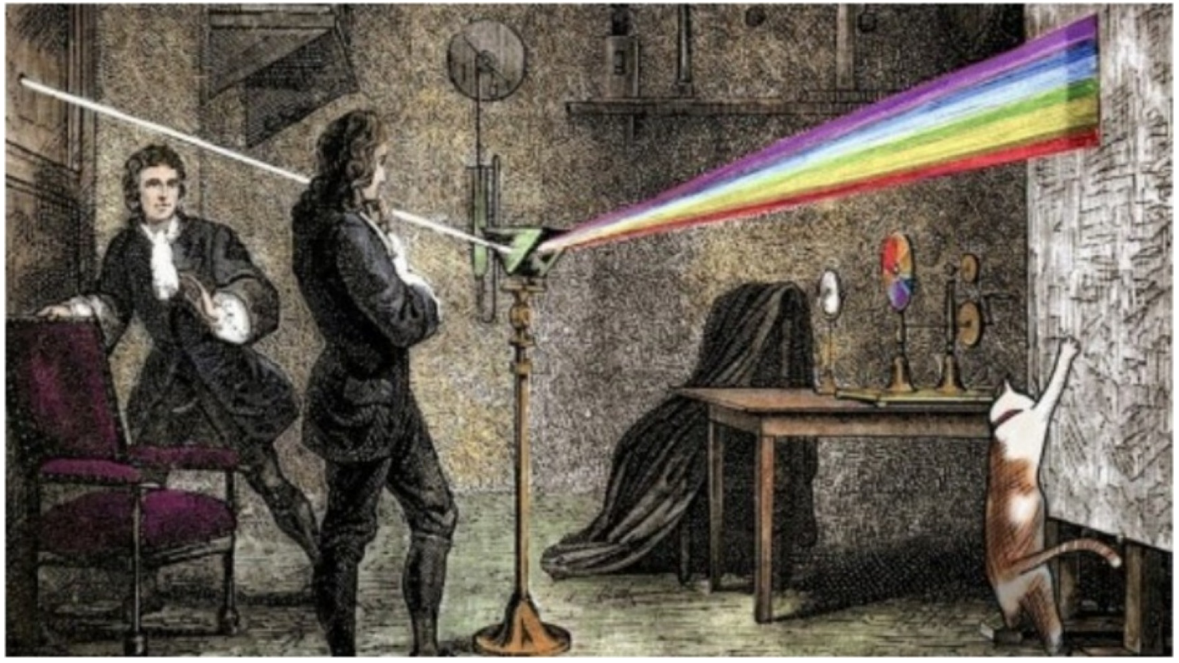
Christoffel equation for plane waves:

$$[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}]\mathbf{a} = \mathbf{0},$$

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l.$$

Definition of the slowness surface:

$$\det[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}] = 0.$$



$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \quad \nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}, \quad \nabla \cdot \mathbf{D} = \rho.$$

Quantum mechanics

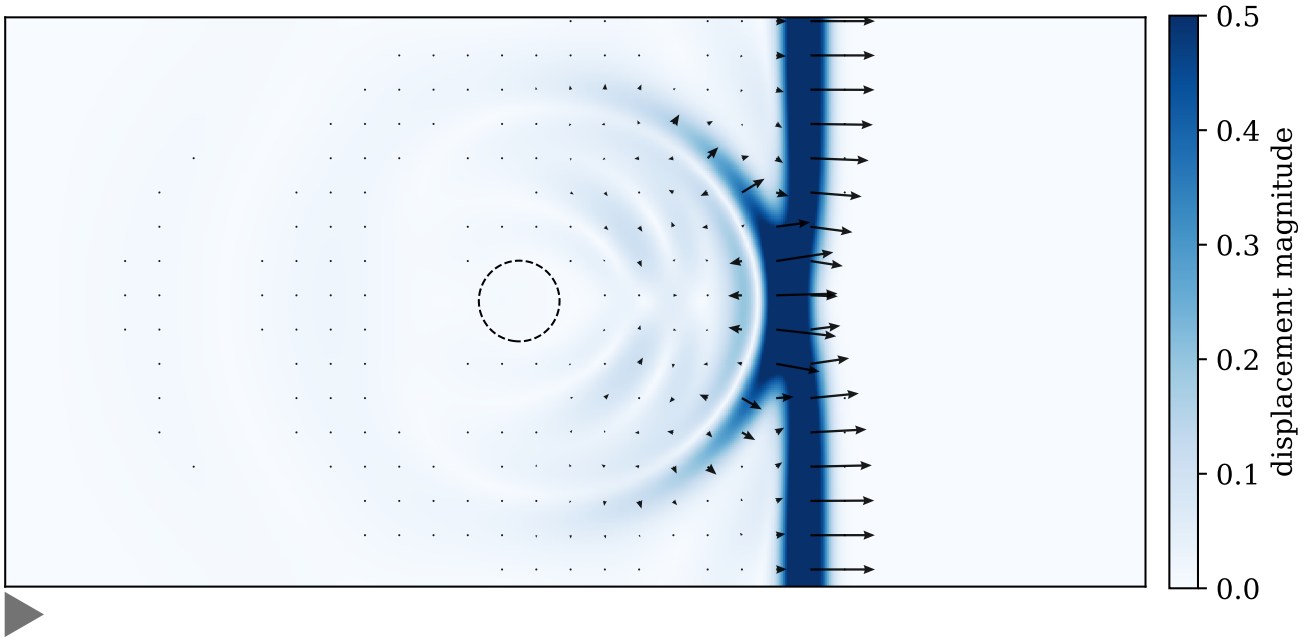
$$\mathrm{i}\hbar\frac{\partial\Psi}{\partial t} = H\Psi$$

Classical mechanics

$$\frac{\mathrm{d}\mathbf{q}}{\mathrm{d}t} = \frac{\partial H}{\partial \mathbf{p}},$$

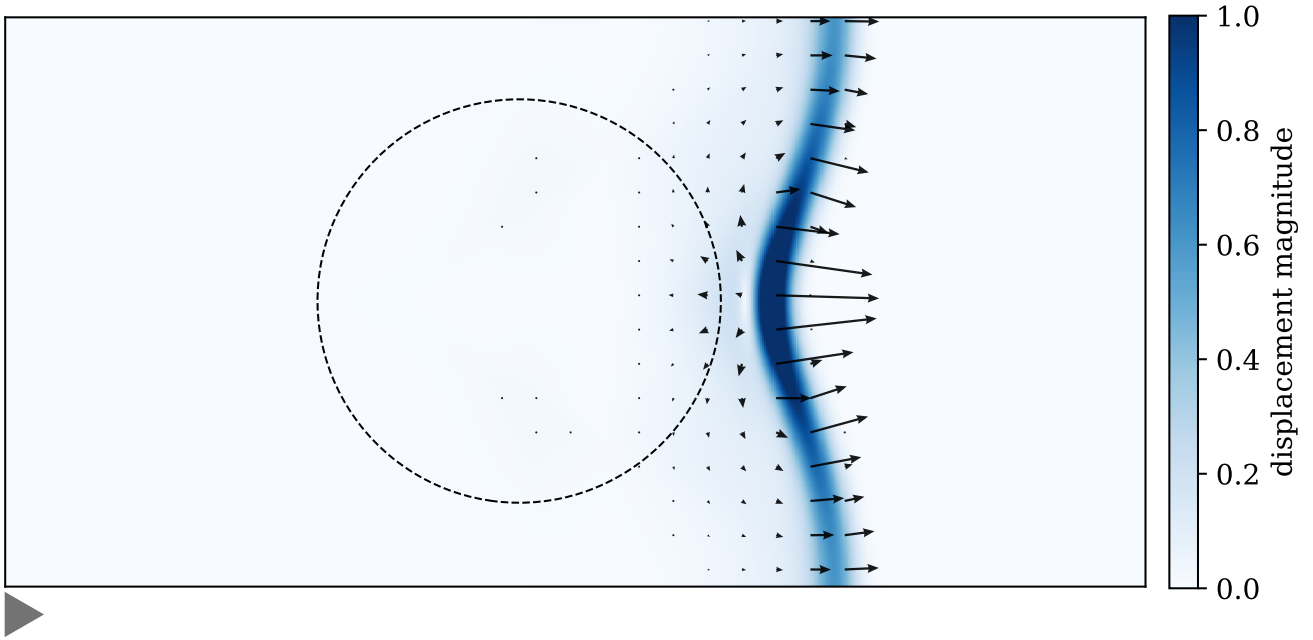
$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = -\frac{\partial H}{\partial \mathbf{q}}$$

Length scales of heterogeneity



Scattering regime, with λ the dominant wavelength and L the length scale of the heterogeneity:

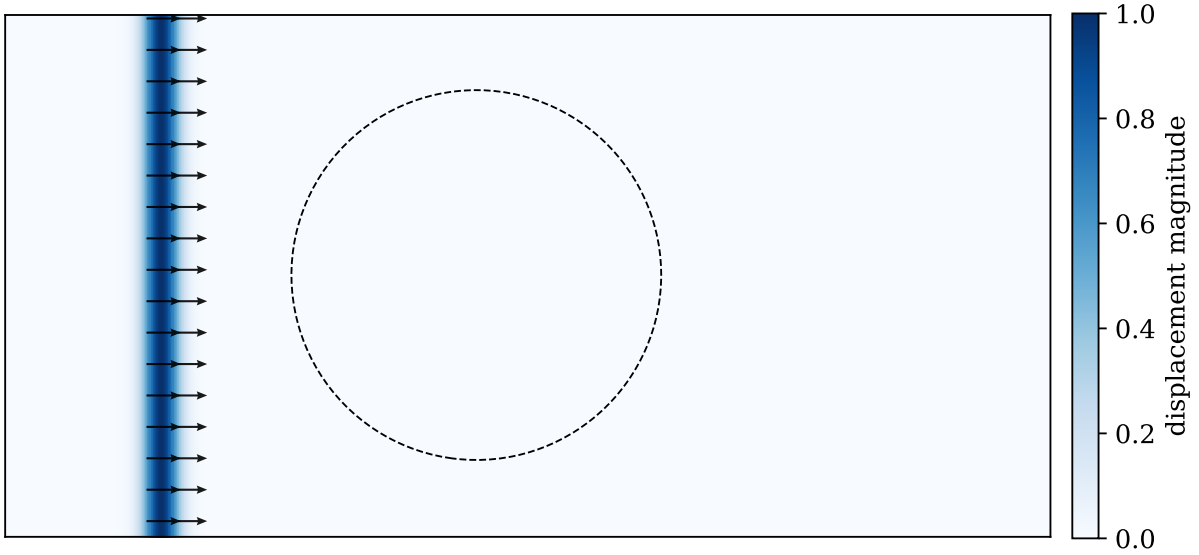
$$\frac{L}{\lambda} \approx 1.$$



Geometric optics regime:

$$\frac{L}{\lambda} \gg 1.$$

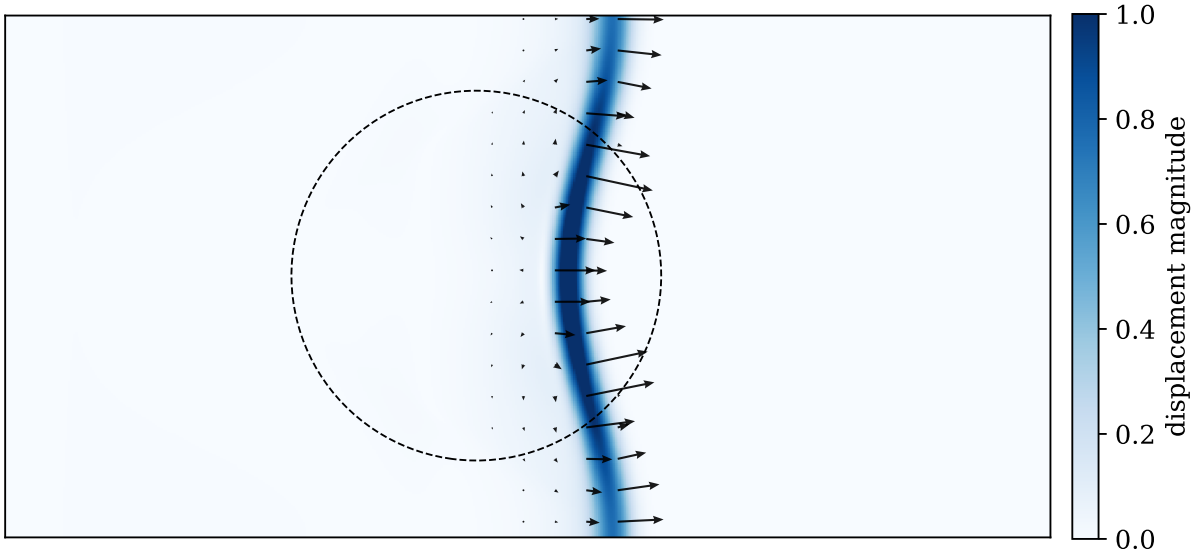
Ray series ansatz



Start with a plane wave within the homogeneous left half-space:

$$u_i(\mathbf{x}, t) = f(t - p_j^0 x_j) a_i^0.$$

Can we extend this solution into the heterogeneous right half-space?



As an initial guess, we look for a solution of the form:

$$u_i(\mathbf{x}, t) = f[t - T(\mathbf{x})]a_i(\mathbf{x}).$$

Assumed solution takes the form:

$$u_i(\mathbf{x}, t) = f[t - T(\mathbf{x})]a_i(\mathbf{x}).$$

Introduction of the **travel time** function allows for curved wavefronts:

$$T(\mathbf{x}) = \text{constant}.$$

The **local slowness vector** is orthogonal to the wavefronts:

$$p_i = \frac{\partial T}{\partial x_i}.$$

Polarisation vector allowed to vary with position – accounts for **focusing effects** and for variations in anisotropic properties with position.

Harmonic decomposition solution:

$$u_i(\mathbf{x},t) = \frac{1}{2\pi}\int_{\mathbb{R}} \tilde{f}(\omega)\mathrm{e}^{\mathrm{i}\omega[t-T(\mathbf{x})]}a_i(\mathbf{x})\,\mathrm{d}\omega.$$

This is a convolution of f with the singular solution:

$$u_i(\mathbf{x},t) = \delta\big[t-T(\mathbf{x})\big]a_i(\mathbf{x}).$$

Experience has shown that we should generalise the singular part to:

$$u_i(\mathbf{x},t) = \frac{1}{2\pi}\int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{a_i^n(\mathbf{x})}{(\mathrm{i}\omega)^n}\mathrm{e}^{\mathrm{i}\omega[t-T(\mathbf{x})]}\,\mathrm{d}\omega.$$

First term ($n=0$) in the **elastodynamic ray series** is the singular one, while successive ones become smoother and smoother.

The eikonal equation

Linearised equations with no force term:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0.$$

Transforming to the frequency domain:

$$-\omega^2 \rho u_i - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0.$$

Ray series in the frequency domain:

$$u_i(\mathbf{x}, \omega) = \sum_{n=0}^{\infty} \frac{a_i^n(\mathbf{x})}{(\mathrm{i}\omega)^n} \mathrm{e}^{-\mathrm{i}\omega T(\mathbf{x})}.$$

We can now substitute in our ansatz and see what happens!

Example step from the calculation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - \mathrm{i}\omega p_l a_k^n \right) \mathrm{e}^{-\mathrm{i}\omega T}.$$

A factor of frequency is added each time we differentiate the phase term.

Equations of motion imply:

$$\sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left[\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - \mathrm{i}\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right.$$

$$\left. - \mathrm{i}\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (\mathrm{i}\omega)^2 A_{ijkl} p_j p_l a_k^n - (\mathrm{i}\omega)^2 \rho a_i^n \right] \mathrm{e}^{-\mathrm{i}\omega T} = 0.$$

Common phase term cancels.

These equations must hold for **all frequencies** – this implies that the **coefficients of each power of frequency must separately vanish**.

Compare with power series solution of ODEs.

From the highest-order coefficient, we have:

$$[A_{ijkl}p_j p_l - \rho\delta_{ik}]a_k^0 = 0.$$

At the next order:

$$[A_{ijkl}p_j p_l - \rho\delta_{ik}]a_k^1 = \frac{\partial}{\partial x_j}(A_{ijkl}p_l a_k^0) + A_{ijkl}p_j\frac{\partial a_k^0}{\partial x_l}.$$

For the general term with $n \geq 2$:

$$\begin{aligned}[A_{ijkl}p_j p_l - \rho\delta_{ik}]a_k^n = \frac{\partial}{\partial x_j}(A_{ijkl}p_l a_k^{n-1}) + A_{ijkl}p_j\frac{\partial a_k^{n-1}}{\partial x_l} \\ - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial a_k^{n-2}}{\partial x_l}\right).\end{aligned}$$

First equation determines $\mathbf{p}$, while later ones successively determine the polarisation vectors at each order.

Where have we made an approximation?

In this course we will consider only the equation derived from the **highest-order** term in the expansion.

If that is all that we require, we can systematically drop all but the highest-order terms within the previous derivation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - \mathrm{i}\omega p_l a_k^n \right) \mathrm{e}^{-\mathrm{i}\omega T} \approx -\mathrm{i}\omega p_l a_k^0 \mathrm{e}^{-\mathrm{i}\omega T}.$$

This saves a lot of time!

You can practise these steps in the first problem set.

From the highest-order coefficient, we have:

$$[A_{ijkl}p_j p_l - \rho\delta_{ik}]a_k^0 = 0.$$

This motivates the definition of a **local Christoffel matrix**:

$$\Gamma_{ik}(\mathbf{x},\mathbf{p}) = \frac{1}{\rho(\mathbf{x})}A_{ijkl}(\mathbf{x})p_j p_l.$$

And so we can write the above condition as:

$$\{\boldsymbol{\Gamma}[\mathbf{x},\mathbf{p}(\mathbf{x})] - \mathbf{I}\}\mathbf{a}^0(\mathbf{x}) = \mathbf{0}.$$

Locally, things look identical to plane waves!

The equation

$$\{\mathbf{\Gamma}[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\}\mathbf{a}^0(\mathbf{x}) = \mathbf{0}$$

has non-trivial solutions for the zeroth-order polarisation vector if and only if:

$$\det\{\mathbf{\Gamma}[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} = 0.$$

This is a non-linear first-order PDE for the travel time called the **eikonal equation**.

If the local slowness surfaces are everywhere non-degenerate:

$$\|\mathbf{p}(\mathbf{x})\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0.$$

The method of characteristics

Recall the eikonal equation is:

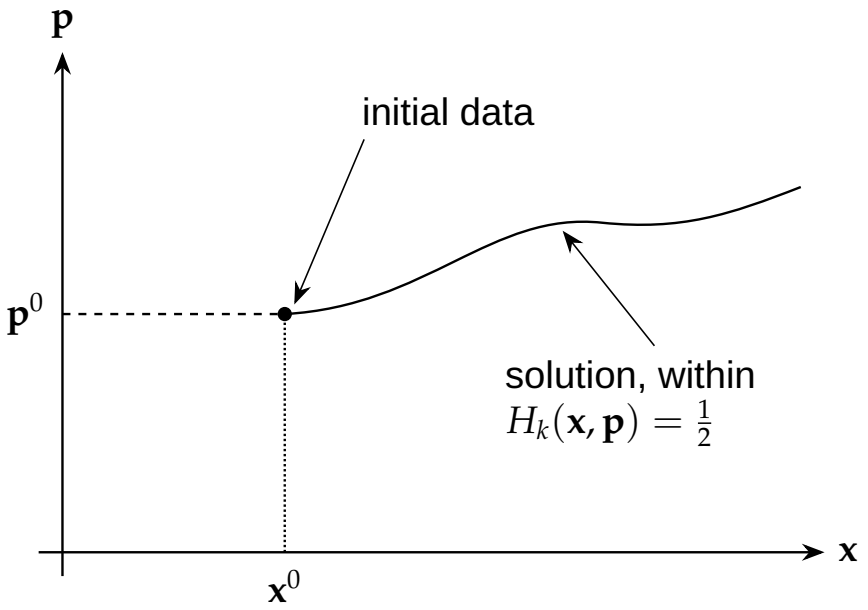
$$\|\mathbf{p}(\mathbf{x})\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0.$$

If we define the **ray Hamiltonian** for the kth sheet of the slowness surface:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|^2 c_k(\mathbf{x}, \hat{\mathbf{p}})^2,$$

we can write the eikonal equation in the form:

$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$



Solution of the eikonal equation is a 3D surface lying within a level set of the ray Hamiltonian in a 6D phase space.

Initial data specifies a 2D surface within this level set.

The problem is how to extend this 2D surface into the appropriate 3D surface.

This is where Hamilton's equations enter into the method.

Initial data extended through integration of Hamilton's equations, with σ a **generating parameter**:

$$\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{\mathrm{d}p_i}{\mathrm{d}\sigma} = -\frac{\partial H_k}{\partial x_i},$$

$$x_i(0) = x_i^0, \quad p_i(0) = p_i^0.$$

This works because the Hamiltonian is constant along the integral curves:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\sigma} H_k[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] &= \frac{\partial H_k}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} + \frac{\partial H_k}{\partial p_i}\frac{\mathrm{d}p_i}{\mathrm{d}\sigma} \\ &= -\frac{\mathrm{d}p_i}{\mathrm{d}\sigma}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} + \frac{\mathrm{d}x_i}{\mathrm{d}\sigma}\frac{\mathrm{d}p_i}{\mathrm{d}\sigma} = 0.\end{aligned}$$

There are a few subtler points to the method that are being ignored.

Rays are the projection of integral curves into the position domain.

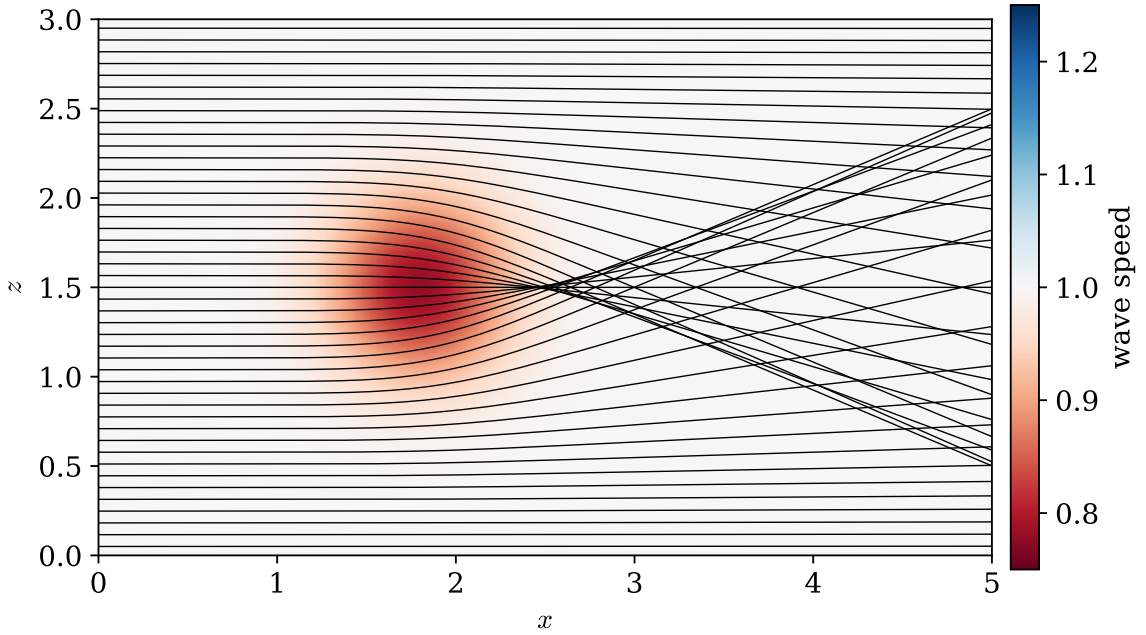
Rate of change of the travel time along a ray:

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} T[\mathbf{x}(\sigma)] = \frac{\partial T}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = p_i \frac{\partial H_k}{\partial p_i}.$$

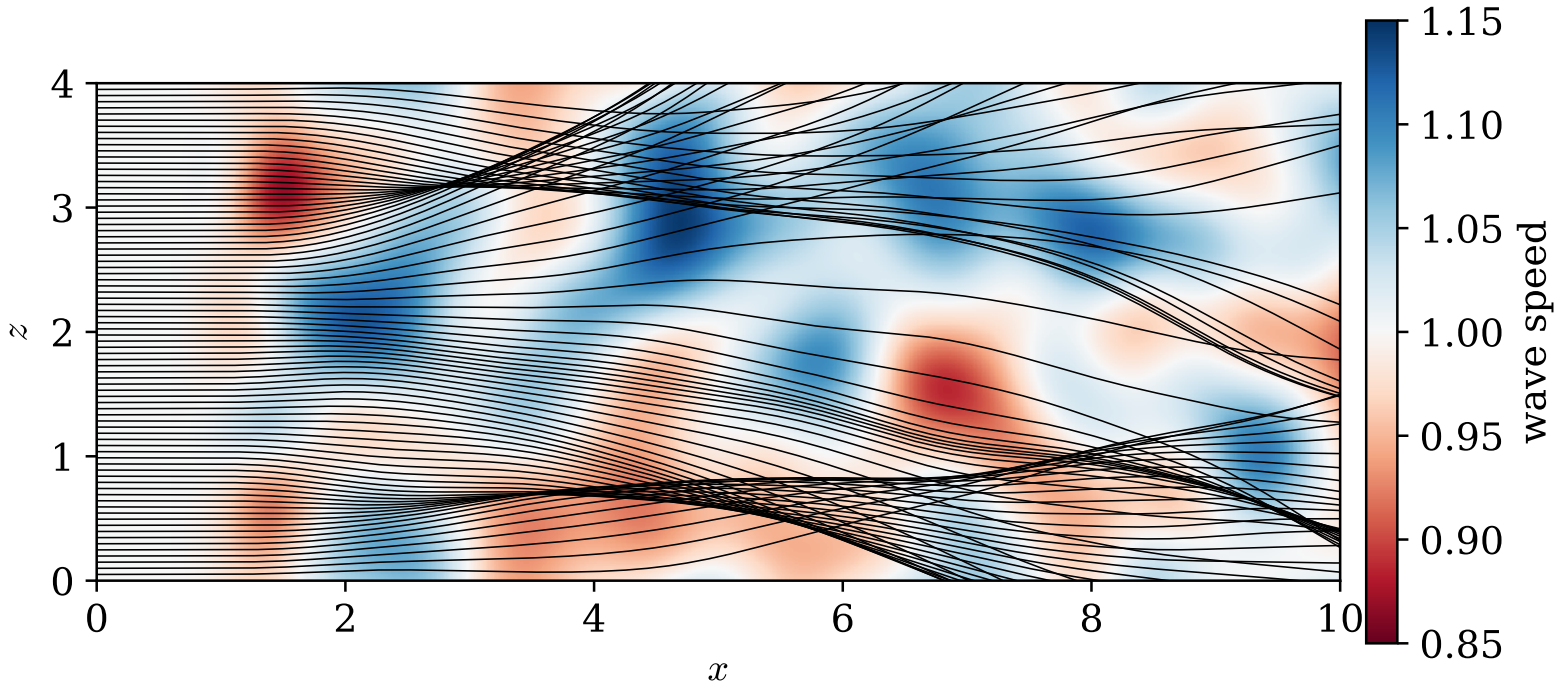
Right-hand side simplified using Euler's theorem for homogeneous functions (proof in the handout, non-examinable):

$$p_i \frac{\partial H_k}{\partial p_i} = 2H_k \quad \longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\sigma} T[\mathbf{x}(\sigma)] = 1.$$

The generating parameter along the ray is equal to the travel time.



If rays for two different integral curves cross, then the travel time need not be well defined. The envelope of the ray family is known as a **caustic**. The **ray theoretic amplitude is infinite** there, while the true amplitude is large but finite.



Within heterogeneous materials, caustics occur frequently – cf. Penrose–Hawking singularity theorems in general relativity.

What you need to know and be able to do

(i) The form of the ray-series expansion, and understand its underlying physical motivation.

(ii) How to obtain the eikonal equation and transport equations by substituting the ray expansion into the equations of motion. The full derivation given in the lecture is too involved to be set in an exam, but you may be asked to apply these methods within simpler situations.

(iii) Understand how the eikonal equation can be interpreted geometrically within a six-dimensional phase space, and be able to explain how solutions can be obtained through integration of Hamilton's equations.

(iv) Be able to solve the Hamiltonian ray equations in simple cases, and also to deduce some of their properties. Practice in this will be given in the next lecture and within the first problem set.

(v) What a caustic is, and why ray theory breaks down there.